VIT-AP UNIVERSITY · SCHOOL OF COMPUTER SCIENCE & ENGINEERING

Hybrid Cloud and Distributed Ledger E-Voting System

Project-Based Component (PBC) Capstone Defense · Academic Year 2025–2026

Project Guide: Dr. Nagendra Panini Challa, Associate Professor, SCOPE

PANEL DEFENDER

Speaker 1 (00–07m)

- Student-1
- 22BCE8319 / 24MIC7027
- Problem & Literature

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Speaker 2 (07–14m)

- Student-2
- 22BEC8319 / 24MIC7307
- Architecture & aBFT

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Speaker 3 (14–22m)

- Student-3
- 22BCE8506 / 24MIC7302
- Concurrency & Live Demo

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Speaker 4 (22–29m)

- Student-4
- 22BCE8947 / 24MIC7348
- Benchmarks & Defense

The Centralized Database Dilemma in Institutional Elections

Speaker 1 · Min 00:00–07:00

VULNERABILITY 01

Privileged DBA Tampering

- Unrestricted Write Access: Root database administrators can alter vote counts or timestamps directly in SQL/NoSQL stores.
- Zero Immutable Audit Trail: Database mutation logs can be purged, truncated, or forged without external detection.
- Internal Insider Threat: Eliminates the guarantee that final tallies represent cast ballots.

VULNERABILITY 02

Black-Box Tallying

- Opaque State Computation: Voters submit credentials and receive an unprovable 'Vote Recorded' acknowledgment.
- Zero Independent Auditing: Outside faculty, observers, or students cannot independently verify that their ballot was counted.
- Erosion of Institutional Trust: Disputed elections cause significant governance and administrative friction.

VULNERABILITY 03

Repudiation & Dispute Friction

- Disenfranchisement Claims: A voter can falsely claim the system failed, or an administrator can deny receipt of a valid vote.
- Lack of Cryptographic Non-Repudiation: No deterministic, tamper-proof receipt exists to resolve disputes conclusively.
- Single Point of Failure: Centralized cloud VM outages halt the entire electoral intake window.

Formulating Trust: Four Core Architectural Objectives

Speaker 1 · Min 00:00–07:00

RESEARCH QUESTION 01

RQ1: Concurrency & Race-Condition Freedom

- How can an off-chain gateway prevent double-voting under concurrent microsecond burst traffic without distributed locking bottlenecks?

RESEARCH QUESTION 02

RQ2: Cost Predictability at Scale

- How can decentralized consensus be integrated into an institutional voting platform while avoiding unpredictable cryptocurrency gas spikes?

RESEARCH QUESTION 03

RQ3: Zero-PII Voter Anonymity

- Can a public distributed ledger achieve 100% public auditability while guaranteeing zero exposure of student personally identifiable information?

RESEARCH QUESTION 04

RQ4: aBFT Consensus Guarantees

- What distributed ledger consensus algorithm guarantees deterministic finality with zero fork or chain reorganization vulnerability?

Architectural Comparison: Hedera HCS vs. Classical Blockchains

Speaker 1 · Min 00:00–07:00

Evaluation Metric	Ethereum L1	Polygon PoS (L2)	Solana	Hedera HCS (Our System)
Consensus Protocol	Proof of Stake (PoS)	PoS Rollup	Proof of History (PoH)	Hashgraph aBFT (Virtual Voting)
Finality Latency	12 – 15 minutes	2 – 4 minutes	400 ms – 12 seconds	3.8 seconds (Deterministic)
Transaction Cost	$2.50 – $45.00 (Gas)	$0.02 – $0.15	$0.0025	$0.0001 USD (Fixed Fiat Peg)
Forking Vulnerability	Probabilistic Forks	Possible Reorgs	Network Outages (History)	Mathematically Immune (aBFT)
Throughput (TPS)	15 – 30 TPS	~7,000 TPS	2,000 – 3,000 TPS	10,000+ TPS Tested
Ledger Governance	Miner / Validator Pool	Multisig Validators	Validator Cluster	Decentralized Governing Council

SYSTEM ARCHITECTURE & FLOW

Decoupled Off-Chain Ingestion & On-Chain Consensus Pipeline

Speaker 2 · Min 07:00–14:00

TIER 1: FRONTEND

1. Client Web Layer

- React 18 Single-Page Application.
- Role-Based Authentication: Voter vs Administrator portals.
- Client-Side Validation: Real-time ballot validation before network dispatch.
- Session Storage: Ephemeral token and receipt storage.

TIER 2: GATEWAY

2. API Gateway

- Node.js & Express REST Service.
- Stateless JWT Verification with tamper-proof signing.
- Express Rate-Limiting: 100 requests / 15 min per IP window.
- CORS & Helmet Security Headers enforcing strict transport isolation.

TIER 3: DATABASE

3. Off-Chain Store

- MongoDB with WiredTiger Storage Engine.
- Document-Level Locking: Exclusive write lock per student record.
- Atomic findOneAndUpdate() eliminates double-voting races.
- PII Isolation: Stores student credentials off-chain.

TIER 4: LEDGER

4. Consensus Layer

- Hedera Consensus Service (HCS).
- Public Topic: 0.0.10697996.
- Consensus Submit Message SDK calls over gRPC TLS.
- Immutable sequence numbering and nanosecond consensus timestamps.

Virtual Voting & Asynchronous Byzantine Fault Tolerance (aBFT)

Speaker 2 · Min 07:00–14:00

MECHANISM 01

Gossip-about-Gossip

- Node Communication: Nodes do not just transmit transactions; they transmit a 2-hash event containing the history of gossip.
- Directed Acyclic Graph (DAG): The graph of events grows exponentially without network bottlenecks.
- Extreme Bandwidth Efficiency: Near-zero consensus messaging overhead compared to Raft or PBFT.

MECHANISM 02

Virtual Voting Algorithm

- Zero Network Overhead: Nodes determine consensus ordering entirely in local memory by inspecting their local DAG view.
- No Extra Vote Messages: Traditional consensus protocols flood the network with vote ballots; Hashgraph votes virtually.
- Fairness of Ordering: Median timestamp consensus prevents miner front-running and ballot reordering attacks.

SECURITY PROOF

Mathematical aBFT Guarantee

- Gold Standard Security: Immune to arbitrary malicious Byzantine nodes and coordinated DDoS attacks.
- Quorum Formulation: Tolerates up to f malicious nodes where n >= 3f + 1.
- Deterministic Finality: Zero probability of blockchain reorganizations, soft forks, or orphan blocks.

PRIVACY PRESERVING ARCHITECTURE

Structural Decoupling: Zero PII Exposure on the Public Ledger

Speaker 2 · Min 07:00–14:00

PRIVATE INSTITUTIONAL BOUNDARY

Off-Chain Private Voter Record (MongoDB)

- Document Content: { _id, voterId: '22BCE8319', email: 'student@vitap.ac.in', hasVoted: true, timestamp }
- Role: Enforces local eligibility, enrollment verification, and one-vote-per-student compliance.
- Access Boundary: Protected behind institutional JWT authentication and strict VPC firewall policies.

PUBLIC CONSENSUS TOPIC 0.0.10697996

On-Chain Public Consensus Message (Hedera HCS)

- Payload Dispatched: { candidateId: 2, timestamp: 1718000000.123 }
- Public HashScan Visibility: Auditors observe that a valid vote was cast, its exact nanosecond, and sequence number.
- Zero Cryptographic Link: Contains zero voter names, student IDs, IP addresses, or database primary keys.

0 Bytes

PII EXPOSED ON-CHAIN

Zero student personal information stored on the public ledger.

100%

INDEPENDENT AUDITABILITY

Any third party can recount the election results directly from Mirror Nodes.

SHA-384

RUNNING HASH INTEGRITY

Each message is chained into an unbroken cryptographic hash sequence.

ECONOMIC VIABILITY & SUSTAINABILITY

Fiat-Pegged Pricing Eliminates Gas Price Volatility

Speaker 2 · Min 07:00–14:00

$0.0001

FIXED HCS TRANSACTION FEE

Directly pegged to USD by the Hedera Governing Council; paid in HBAR.

$1.00

TOTAL COST / 10,000 VOTES

Entire university elections can be conducted with negligible financial overhead.

98.4%

COST REDUCTION VS. POLYGON

Immune to network congestion, MEV bot bidding wars, and gas surges.

INSTITUTIONAL BUDGET GUARANTEE

Economic Model Comparison & Budget Predictability

- Ethereum L1 Vulnerability: Gas surges during high DeFi volume can push ballot submission costs above $15.00 per vote, breaking institutional budgets.
- Polygon & Rollup Volatility: Even Layer-2 gas fees fluctuate wildly during network congestion, requiring complex gas treasury auto-replenishment systems.
- Hedera Governing Council Stability: Fees are fixed in fiat USD and automatically converted to HBAR at execution time. A 15,000-student election has a guaranteed maximum cost of exactly $1.50 USD.

Preventing Race Conditions via WiredTiger Document Locking

0.0%

DOUBLE-VOTING MARGIN

Zero double-votes recorded across 500 concurrent burst tests.

100%

WRITE SERIALIZATION

MongoDB WiredTiger document-level locking serializes concurrent updates.

403

FORBIDDEN STATUS

Second concurrent thread instantly rejected before reaching Hedera.

$ concurrency_controller.js

```
// Atomic Claim in Node.js / Express Gateway
const voter = await Voter.findOneAndUpdate(
  { voterId: req.user.voterId, hasVoted: false }, // Atomic Filter
  { $set: { hasVoted: true, votedAt: new Date() } },
  { new: true, session } // Document-level write lock
);

if (!voter) {
  // 2nd concurrent thread evaluates to null
  return res.status(403).json({
    error: 'Ballot already cast or claim in progress.'
  });
}
// Proceed safely to Hedera HCS submission...
```

FAULT TOLERANCE & STATE RECOVERY

Fail-Safe Atomic Rollback Prevents Voter Disenfranchisement

Speaker 3 · Min 14:00–22:00

PHASE 1: ATOMIC CLAIM

Phase 1: Local State Lock

- Voter dispatches authenticated HTTP POST /vote request.
- MongoDB executes atomic findOneAndUpdate({ hasVoted: false }).
- State immediately marked hasVoted: true in local memory.
- Ensures no concurrent duplicate request can enter the pipeline while HCS network call is pending.

PHASE 2: CONSENSUS ORDERING

Phase 2: Remote HCS Consensus

- Backend constructs Hedera TopicMessageSubmitTransaction.
- Payload signed with institutional Operator Key and sent over gRPC TLS.
- Hedera Hashgraph consensus network assigns immutable timestamp and sequence number.
- Receipt transaction ID returned to caller.

FAIL-SAFE RECOVERY

Fail-Safe Rollback Handler

- Catastrophic failure handling: network partition or Hedera gRPC timeout.
- Automated catch block detects missing consensus receipt.
- Executes atomic rollback: $set { hasVoted: false }.
- Voter is NOT disenfranchised and can safely retry ballot submission.

SYSTEM DEMONSTRATION

Live System Verification: End-to-End Ballot Journey

Speaker 3 · Min 14:00–22:00

DEMO STAGE 1: FRONTEND

Step 1: Voter Portal Cast

- Voter logs into React SPA (http://localhost:5173).
- Selects Candidate and clicks 'Submit Vote'.
- Real-time visual feedback: Button enters loading state, dispatches signed payload to port 5000.
- Browser displays instant success dialog with cryptographic Transaction ID.

DEMO STAGE 2: HASHSCAN

Step 2: HashScan Explorer

- Switch to Browser Tab 2: HashScan Explorer.
- Inspect public topic: https://hashscan.io/testnet/topic/0.0.10697996.
- Observer verifies: Message count increments by 1 in real-time.
- Inspect consensus timestamp, sequence number, and running hash.

DEMO STAGE 3: MIRROR API

Step 3: Mirror Node Audit

- Switch to Browser Tab 3: Mirror Node REST API.
- Query GET /api/v1/topics/0.0.10697996/messages.
- Demonstrate JSON payload decoding: candidateId: 2, timestamp: 1718000000.123.
- Proves independent, trustless auditability without database access.

Trustless Verification: HashScan Explorer & Mirror Node REST API

Speaker 3 · Min 14:00–22:00

AUDIT COMPONENT 01

Cryptographic Transaction ID Receipt

- Receipt Structure: 0.0.10697996@1718000000.123456789.
- Issued directly to the voter upon consensus confirmation.
- Non-Repudiable Evidence: Neither the student nor the administration can falsely claim whether a vote was cast.
- Enables independent ballot tracking across public blockchain explorers worldwide.

AUDIT COMPONENT 02

Decentralized Mirror Node REST API

- Public API Endpoint: mainnet-public.mirrornode.hedera.com.
- Zero Rate-Limits for Civic Auditors: Mirror nodes run independently of consensus nodes.
- Running SHA-384 Hash Chaining: Each message contains the running hash of all previous messages, making history deletion mathematically impossible.
- Automated Verification Scripts: Anyone can tally the election with a 10-line Python or cURL script.

EMPIRICAL RESULTS & BENCHMARKS

System Performance: Latency, Throughput, and Finality Under Load

Speaker 4 · Min 22:00–29:00

3.82s

END-TO-END FINALITY

Measured average time from user click to global Hedera consensus ordering.

68 ms

API GATEWAY LATENCY

Local JWT verification, WiredTiger claim, and payload serialization time.

100.0%

CONSENSUS DELIVERY RATE

Zero dropped messages or consensus timeout failures across all test rounds.

BENCHMARK METHODOLOGY

Benchmark Execution Methodology & Test Conditions

- Test Bed: 500 concurrent ballot requests simulated via automated test harness under variable network delay.
- Throughput Stability: Hedera Consensus Service maintained constant sub-4-second finality with zero queuing latency or memory leaks.
- Cost Stability: Total network fee across 500 test votes was exactly $0.05 USD, matching theoretical models perfectly.

SECURITY & THREAT ANALYSIS

Comprehensive STRIDE Threat Modeling Evaluation

Speaker 4 · Min 22:00–29:00

STRIDE Threat	Attack Vector	Architectural Mitigation	Residual Risk
Spoofing	Unauthorized voter impersonation	Stateless cryptographically signed JWTs + password hashing	Negligible (Protected by Auth)
Tampering	Altering ballot tally in database	Public Hedera HCS sequence numbering + SHA-384 running hash	Zero (Tamper-Evident Ledger)
Repudiation	Voter denies casting ballot	Deterministic transaction ID receipt verified on HashScan explorer	Zero (Cryptographic Proof)
Information Disclosure	Eavesdropping voter identities	Zero PII stored on ledger; structural decoupling of ballot payload	Zero (Decoupled Payload)
Denial of Service	Gateway flood attacks	Express rate limiting (100 req/15m) + stateless gateway architecture	Low (Scalable via Cloud Load Balancer)
Elevation of Privilege	Student escalates to admin	Role-based authorization middleware enforcing strict permission gates	Negligible (Strict RBAC)

Architectural Evolution: zk-SNARKs & Threshold Multi-Sig Governance

Speaker 4 · Min 22:00–29:00

MILESTONE 1: GOVERNANCE

Threshold Multi-Sig Keys

- Current Limitation: Testnet uses a single institutional operator key to submit HCS messages.
- Roadmap Solution: Deploy Hedera multi-signature threshold accounts on mainnet.
- Governance Guardrail: Requires 3 out of 5 designated faculty election commissioners to co-sign the submission transaction.

MILESTONE 2: PRIVACY

Client-Side zk-SNARKs

- Current Limitation: Database-level decoupling prevents public observer tracking, but a compromised DBA could correlate database timestamps.
- Roadmap Solution: Implement Zero-Knowledge Succinct Non-Interactive Arguments of Knowledge (zk-SNARKs).
- Mathematical Anonymity: Voter proves eligibility cryptographically in the browser with 0 correlation to cast ballot.

MILESTONE 3: IDENTITY

Decentralized Identity (DID)

- Current Limitation: Relies on institutional username/password authentication.
- Roadmap Solution: Integrate W3C Verifiable Credentials and self-sovereign Decentralized Identifiers (DIDs).
- Decentralized Login: Students authenticate directly via digital identity wallets without centralized institutional ID databases.

Summary of Contributions & Open Technical Defense

All Team Members · Evaluator Defense

DELIVERED CONTRIBUTIONS

Key Technical Contributions

- Decoupled Hybrid Architecture: Combines local database throughput with global Hedera Hashgraph aBFT consensus integrity.
- Sub-Second Finality: 3.82s deterministic finality with zero blockchain reorganization risk.
- Cost Sustainability: $0.0001 USD fixed transaction fee eliminates gas volatility.
- Race-Condition Freedom: Proven WiredTiger document-level locking eliminates double-voting entirely.
- Public Trustless Verification: 100% independent tally auditability via HashScan and public Mirror Node REST APIs.

PANEL DEFENSE READINESS

Evaluator Defense Focus Areas

- Q1: Operator Key Single Point of Failure -> Separation of submission rights from consensus ordering; Threshold Multi-sig migration.
- Q2: Race Condition Prevention -> MongoDB WiredTiger exclusive write locking on { voterId, hasVoted: false }.
- Q3: HCS vs Ethereum Rollups -> Fiat-pegged fee predictability ($1/10k votes) vs gas volatility; sub-4s deterministic aBFT finality.
- Q4: Voter Identity Privacy -> Structural decoupling; payload contains { candidateId, timestamp } with zero student PII.
- Q5: Non-Repudiation Proof -> Cryptographic Transaction IDs verified on public HashScan explorer.